MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

MANIPAL UNIVERSITY JAIPUR

Department of Computer Science and Engineering (IoT and Intelligent Systems)
Course Hand-out

A. Basic Details:

Programme Name:	B.Tech CSE (IoT and Intelligent Systems)
Course Name:	IoT Forensics
Course Code:	IIS3242
LTPC *(Lecture Tutorial Practical Credits):*	(3003)
Session:	Jan 2026 – May 2026
Class:	VI Semester
Course Coordinator:	Dr. Rajesh Kumar
Course Instructor(s):	Dr. Rajesh Kumar
Additional Practitioner(s) – *(Industry Fellow, Subject to approval from concerned authority)*	

B. Introduction: This course introduces the fundamentals of IoT Forensics, focusing on how digital investigations are carried out in IoT environments. It covers forensic procedures, evidence acquisition, secure preservation, and analysis of data from devices, networks, cloud, and edge systems. Students will learn about modern forensic tools, email and network investigations, and legal aspects of forensics. The course also highlights the use of machine learning and the importance of maintaining chain of custody in IoT forensic investigations.

C. Course Outcomes:

At the end of the course, students will be able to:

Explain the fundamentals of digital forensics, including benefits, vulnerabilities, threats, and legal considerations in IoT environments. (Understand – Bloom's Level: L2)

Apply standard procedures and tools for corporate digital investigations, including data recovery workstations, data acquisition methods, and secure evidence preservation. (Apply – Bloom's Level: L3)

Analyze digital evidence from IoT systems using network, cloud, edge, and email forensics techniques while validating and testing forensic tools. (Analyze – Bloom's Level: L4)

Design and evaluate IoT forensic investigation workflows using machine learning techniques while maintaining proper chain of custody. (Evaluate/Create – Bloom's Level: L5–L6)

CO Statement	*CO*	*Level*	*Target Attainment %*	*Target Attainment level*
Explain the fundamentals of digital forensics, including benefits, vulnerabilities, threats, and legal considerations in IoT environments.	IIS3242.1	2	80	3
Apply standard procedures and tools for corporate digital investigations, including data recovery workstations, data acquisition methods, and secure evidence preservation.	IIS3242.2	2	75	2
Analyze digital evidence from IoT systems using network, cloud, edge, and email forensics techniques while validating and testing forensic tools.	IIS3242.3	3	75	2
Design and evaluate IoT forensic investigation workflows using machine learning techniques while maintaining proper chain of custody.	IIS3242.4	3	70	2

Here, the level indicates the cognitive level of Revised Bloom's Taxonomy. Information about attainment levels:

Attainment (%)	**Level**
< 60 %	0
≥ 60% < 70%	1
≥ 70% < 80%	2
≥ 80	3

D. Program Outcomes and Program Specific Outcomes:

[PO. 1] Engineering knowledge: apply the knowledge of mathematics, computer science, and communication engineering fundamentals to the solution of complex engineering problems.
[PO. 2] Problem analysis: identify, formulate, review research literature, and analyse complex engineering problems reaching substantiated conclusions using basic principles of mathematics, computing techniques and communication engineering principles.
[PO. 3] Design/development of solutions: design solutions for complex engineering problems and design system components or processes that meet the specified requirements with appropriate consideration for law, safety, cultural & societal obligations with environmental considerations.
[PO. 4] Conduct investigations of complex problems: use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
[PO. 5] Modern tool usage: create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modelling to complex engineering activities with an understanding of the limitations.
[PO. 6] The engineer and society: apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
[PO. 7] Environment and sustainability: understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

[PO. 8] Ethics: apply ethical principles and commit to professional ethics, responsibilities and norms of the engineering practice.
[PO. 9] Individual and teamwork: function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
[PO. 10] Communication: communicate effectively for all engineering processes & activities with the peer engineering team, community and with society at large. Clarity of thoughts, being able to comprehend and formulate effective reports and design documentation, make effective presentations, and give and receive clear instructions.
[PO. 11] Project management and finance: demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in varied environments.
[PO. 12] Life-long learning: recognize the need for and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

Program Specific Outcome

[PSO.1] Apply the fundamental knowledge of computer science and engineering in developing effective software/hardware for real world complex engineering problems by adapting advanced technologies.

[PSO.2] Analyze and configure various IoT based innovative and smart applications using recent hardware and software tools.

[PSO.3] Design industrial IoT based solutions for improving operational efficiency at home and industry automation systems.

E. Assessment Plan:

Criteria	Description	Maximum Marks
Internal Assessment (Summative)	Mid-Term Examination (Close Book)	30
	Class Work Sessional (CWS): 1. 10 marks for Class Activities. 2. 10 marks for quizzes, average of three out of four quizzes will be considered. 3. 10 marks for assignments/ video assignment/ viva.	30
End Term Exam (Summative)	End Term Exam (Close Book)	40
	Total	100

F. Syllabus: Introduction to IoT Forensics: Digital forensics, benefits, vulnerabilities and threats in IoT environments. **Understanding Computing Investigations:** Procedure for corporate high-tech investigations, understanding data recovery workstation and software. **Data Acquisition:** Storage formats and digital evidence, selection and optimization of acquisition methods, acquisition tools, validating data acquisitions, secure evidence

preservation in forensics, network forensics, cloud and edge forensics in IoT, legal and compliance issues in forensics. **Current Computer Forensics Tools:** Software, hardware tools, validating and testing forensic software, addressing data hiding techniques, performing remote acquisitions, understanding email servers, investigating email crime and violations, specialized email forensics tools. **Machine Learning Applications:** Forensic investigations, chain of custody in IoT forensics.

References:

1. Anita Gehlot, Rajesh Singh, Jaskaran Singh, Neeta Raj Sharm, Digital Forensics and Internet of Things: Impact and Challenges, 1st edition, Wiley, 2022.
2. Mandia, K. Computer Forensics: A Pocket Guide. O'Reilly Media, 2021.
3. G. Johansen, Digital Forensics and Incident Response, 2nd edition, Packet Publishing, 2020.
4. Nelson, B., Phillips, A., & Steuart, C, Guide to Computer Forensics and Investigations, 7th edition, Cengage Learning, 2019.

G. Lecture Plan:

Lecture	Topics	Session Outcome	Corresponding CO	Mode of delivery	Mode of Assessing CO
1	Introduction of course	Understanding of course, objectives, evaluation	NA	Lecture	NA
2	Introduction to IoT Forensics	What are Digital Forensics	IIS3242.1	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
3		Digital forensics benefits	IIS3242.1	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
4-6		Vulnerabilities and threats in IoT environments	IIS3242.1	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
7-8	Understanding Computing Investigations	Procedure for corporate high-tech investigations	IIS3242.1	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
9-10		Understanding data recovery workstation and software	IIS3242.1	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
11	Data Acquisition	Storage formats	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term

12		Digital evidence	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
13-14		Selection and optimization of acquisition methods	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
15		Data acquisition tools	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
16		Validating the data acquisitions	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
17		Secure evidence preservation in forensics	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
18		Network forensics	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
19-20		Cloud and edge forensics in IoT	IIS3242.2	Peer Teaching	Sessional Exam, Quiz & End Term
21-22		Legal and compliance issues in forensics.	IIS3242.2	Lecture based teaching-learning	Sessional Exam, Quiz & End Term
Mid Term Exam					
23	Current Computer Forensics Tools	Software and hardware tools	IIS3242.3	Lecture based teaching-learning	Quiz & End Term
24-26		Validating and testing forensic software	IIS3242.3	Lecture based teaching-learning	Quiz & End Term
27-28		Addressing data hiding techniques	IIS3242.3	Learning through problem solving	Quiz & End Term
29		Performing remote acquisitions	IIS3242.3	Group Teaching and learning	Quiz & End Term

30		Understanding email servers	IIS3242.3	Group Teaching and learning	Quiz & End Term
31-33		Investigating email crime and violations	IIS3242.3	Lecture based teaching learning	Quiz & End Term
34		Specialized email forensics tools	IIS3242.3	Game based learning	Quiz & End Term
35	Machine Learning Applications	Forensic investigations	IIS3242.4	Lecture based teaching learning	Quiz & End Term
36-38		Chain of custody in IoT forensics.	IIS3242.4	Lecture based teaching learning	Quiz & End Term
End-Term Examination					

H. Outside class engagement activities (Min 72 Hours)

Type of Activity	Difficulty Level	Hours Engaged
Self-Learning Course IoT forensics	High	70 Hours
Quiz / Self-Assessment (Online)	Easy	2 Hours
Self-Study (Books + Web Resources)	Moderate	20 Hours

I. Learner centric activities:

Activities at low-difficulty level:

- One-on-one remedial sessions or tutorials

Activities at high-difficulty level:

- Assignments involving real-life application or case studies
- Mentoring peers or leading study groups

J. Course Articulation Matrix:

CO	STATEMENT	CORRELATION WITH PROGRAM OUTCOMES AND PROGRAM SPECIFIC OUTCOMS														
		PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PO6**	**PO7**	**PO8**	**PO9**	**PO10**	**PO11**	**PO12**	**PSO1**	**PSO2**	**PSO3**
CO1	Explain the fundamentals of digital forensics, including benefits, vulnerabilities, threats, and legal considerations in IoT environments.	3	2	1	1	2	2	1	2		1		2	2	1	
CO2	Apply standard procedures and tools for corporate digital investigations, including data recovery workstations, data acquisition methods, and secure evidence preservation.	3	3	2	2	3	2	1	2		2	1	2	3	2	
CO3	Analyze digital evidence from IoT systems using network, cloud, edge, and email forensics techniques while validating and testing forensic tools.	3	3	2	3	3	2	1	2		2	1	3	3	3	2
CO4	Design and evaluate IoT forensic investigation workflows using machine learning techniques while maintaining proper chain of custody.	3	3	3	3	3	2	1	2	2	2	2	3	3	3	3

1: Low Correlation 2: Moderate Correlation 3: Substantial Correlation